Lathi 20题练习册 第6批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

等能量软启动脉冲的持续时间

Lathi 1.1-5 | 原书第 136 页

Boulet 第1章：Finite-Energy and Finite-Power Signals, p.21

1.1-5 By original design, a system outputs a 10-volt pulse that is 3 seconds in duration. It is desired to upgrade the square-pulse output with a "soft-start" pulse that steps up to 10 volts in 1-volt increments spaced every 20 milliseconds. Determine the signal duration T so that the "soft-start" pulse has the same signal energy as the original square pulse.

演算区

周期、功率与时间缩放性质辨析

Lathi 1.3-4 | 原书第 140 页

Boulet 第1章：Periodic Signals, p.8；Finite-Energy and Finite-Power Signals, p.21

原题同页跨栏，题干与(a)-(d)已完整拼接。

1.3-4 Determine whether each of the following statements is true or false. If the statement is false, demonstrate by proof or example why the statement is false.

(a) Every bounded periodic signal is a power signal.

(b) Every bounded power signal is a periodic signal.

(c) If an energy signal $x(t)$ has energy E, then the energy of $x(at)$ is E/a. Assume a is real and positive.

(d) If a power signal $x(t)$ has power P, then the power of $x(at)$ is P/a. Assume a is real and positive.

演算区

周期拼接信号的变换、微分与能量功率

Lathi 1.4-3 | 原书第 141 页

Boulet 第1章：Transformations of the Time Variable，p.4；Periodic Signals，p.8；Generalized Functions，p.26

1.4-3 Letting $w(t) = t\,[u(t) - u(t-1)]$, define the periodic signal $x(t)$ as

$$x(t) = \sum_{k=-\infty}^{\infty} w(2t+2k) - 0.5w(2t+2k-1)$$

(a) Sketch $w(t)$ and $x(t)$. What is the fundamental period T_0 of signal $x(t)$?
(b) Sketch $y(t) = \frac{d}{dt}x(1-0.5t)$.
(c) Determine the energy E_z and power P_z of the signal $z(t) = x(0.5 - 1.5t)\,[u(t) - u(t-1)]$. Sketching $z(t)$ should help.

演算区

系统可逆性与逆系统关系

Lathi 1.7-16 | 原书第 146 页

Boulet 第1章：System Models and Basic Properties, p.34

1.7-16 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which are invertible and which are noninvertible. For the invertible systems, find the input–output relationship of the inverse system.

(a) $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$

(b) $y(t) = x^n(t)$, $x(t)$ real, n integer

(c) $y(t) = \dfrac{dx(t)}{dt}$

(d) $y(t) = x(3t - 6)$

(e) $y(t) = \cos[x(t)]$

(f) $y(t) = e^{x(t)}$, $x(t)$ real

演算区

离散指数信号与复平面位置

Lathi 3.3-7 | 原书第 317 页

Boulet 第1章：Exponential Signals，p.9；Periodic Complex Exponential and Sinusoidal Signals，p.17

3.3-7 The following signals are in the form $e^{\lambda n}$. Express them in the form γ^n:

(a) $e^{-0.5n}$

(b) $e^{0.5n}$

(c) $e^{-j\pi n}$

(d) $e^{j\pi n}$

In each case show the locations of λ and γ in the complex plane. Verify that an exponential is growing if γ lies outside the unit circle (or if λ lies in the RHP), is decaying if γ lies within the unit circle (or if λ lies in the LHP), and has a constant amplitude if γ lies on the unit circle (or if λ lies on the imaginary axis).

演算区

图解卷积的分段积分与指定时刻值

Lathi 2.4-1 | 原书第 225 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

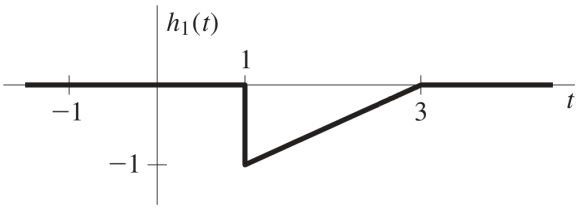
原图 P2.4-1 的两个信号已附；(b)只要求列积分，(c)要求给出数值。

2.4-1 Let $f(t) = h_1(t) * h_2(t)$, where $h_1(t)$ and $h_2(t)$ are shown in Fig. P2.4-1. In the following, use the graphical convolution procedure where you flip and shift $h_2(t)$.

(a) Plot $h_1(\tau)$ and $h_2(t-\tau)$ as functions of τ. Clearly label the plots, including necessary function parameterizations.

(b) Determine the (piecewise) regions of $f(t)$ and set up the corresponding integrals that describe $f(t)$ in those regions. Do not evaluate the integrals, only set them up!

(c) Determine $f(1)$, which is $f(t)$ evaluated at $t = 1$. Provide a number, not a formula.



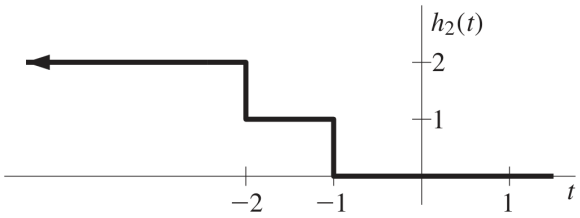


Figure P2.4-1

演算区

卷积输出的支撑终点与极值位置

Lathi 2.4-2 | 原书第 225 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-2 Consider signals $h(t) = u(t+3) - 2u(t+1) + u(t-1)$ and $x(t) = \cos(t)\left[u(t-\pi/2) - u(t-3\pi/2)\right]$. Let $y(t) = x(t) * h(t)$.

(a) Determine the last time t_{last} that $y(t)$ is nonzero. That is, find the smallest value t_{last} such that $y(t) = 0$ for all $t > t_{\text{last}}$.

(b) Determine the approximate time t_{max} where $y(t)$ is a maximum.

演算区

由输入输出反求连续系统冲激响应

Lathi 2.4-23 | 原书第 228 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

2.4-23 The zero-state response of an LTIC system to an input $x(t) = 2e^{-2t}u(t)$ is $y(t) = [4e^{-2t} + 6e^{-3t}]u(t)$. Find the impulse response of the system. [*Hint:* We have not yet developed a method of finding $h(t)$ from the knowledge of the input and the corresponding output. Knowing the form of $x(t)$ and $y(t)$, you will have to make the best guess of the general form of $h(t)$.]

演算区

非因果有限响应系统的周期序列卷积

Lathi 3.8-4 | 原书第 321、322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

原题跨页，冲激响应、完整六周期输入定义及(a)、(b)均已附。

3.8-4 An LTID system has impulse response $h[n] = (0.5)^{(n+3)}\,(u[n] - u[n+6])$. A 6-periodic DT input signal $x[n]$ is given by

$$x[n] = \begin{cases} 1 & n = 0, \pm 3, \pm 6, \pm 9, \pm 12, \ldots \\ 2 & n = 1, 1 \pm 6, 1 \pm 12, \ldots \\ 3 & n = 2, 2 \pm 6, 2 \pm 12, \ldots \\ 0 & \text{otherwise} \end{cases}.$$

(a) Is system $h[n]$ causal? Mathematically justify your answer.

(b) Determine the value of $y[12]$, the zero-state output of system $h[n]$ in response to $x[n]$ at time $n = 12$. Express your result in decimal form to at least three decimal places (e.g., $y[12] = 1.234$).

演算区

离散LTI系统互联及逆系统

Lathi 3.8-18 | 原书第 322、323 页

Boulet 第2章：Properties of Linear Time-Invariant Systems， p.74

所需原图 P3.8-18 已完整附上。

3.8-18 (a) A discrete-time LTI system is shown in Fig. P3.8-18. Express the overall impulse response of the system, $h[n]$, in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$, $h_4[n]$, and $h_5[n]$.

(b) Two LTID systems in cascade have impulse response $h_1[n]$ and $h_2[n]$, respectively. Show that if $h_1[n] = (0.9)^n u[n] - 0.5(0.9)^{n-1}u[n-1]$ and $h_2[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1}u[n-1]$, the cascade system is an identity system.

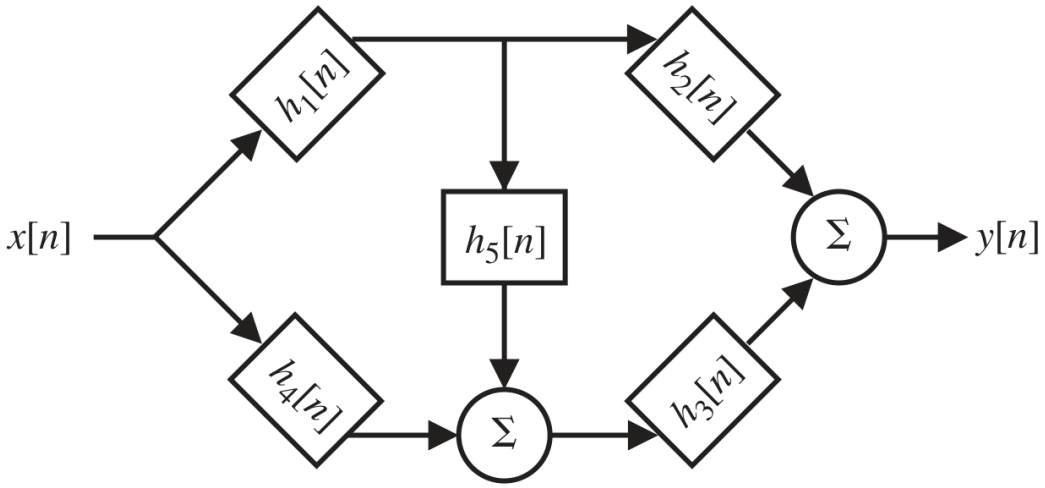


Figure P3.8-18

演算区

含零特征根的连续系统零输入响应

Lathi 2.2-6 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按给定初值求 t>=0 的零输入响应。

2.2-6 Repeat Prob. 2.2-4 for

$$D(D+1)y(t) = (D+2)x(t)$$

and $y_0(0^-) = \dot{y}_0(0^-) = 1.$

演算区

运放RC电路的单位冲激响应

Lathi 2.3-6 | 原书第 225、224 页

Boulet 第3章：Impulse Response of a Differential LTI System，p.101

所引用原图 P2.2-12 已附，元件数值以本题题干为准。

2.3-6 Determine and plot the unit impulse response $h(t)$ of the op-amp circuit of Fig. P2.2-12, assuming that $R = 300\ \text{k}\Omega$, $R_f = 1.2\ \text{M}\Omega$, and $C = 5\ \mu\text{F}$.

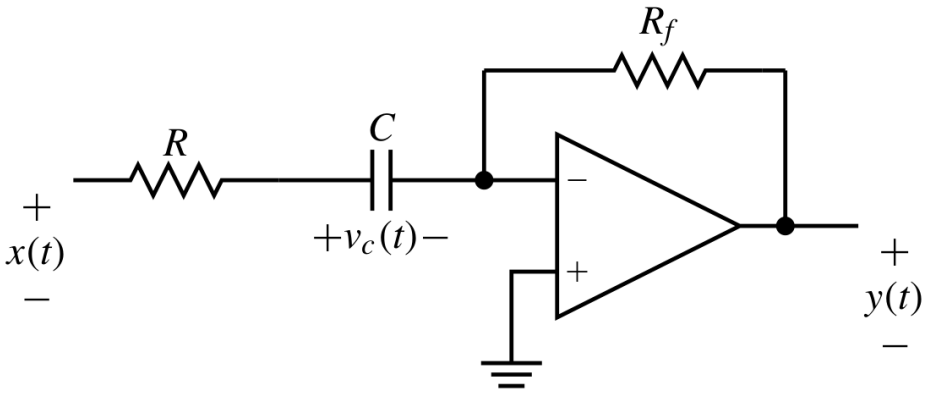


Figure P2.2-12

演算区

二阶齐次差分方程与辅助条件

Lathi 3.6-2 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.6-2 Solve

$$y[n+2]+3y[n+1]+2y[n]=0$$

if $y[-1]=0$ and $y[-2]=1$.

演算区

超前下标差分系统的模态与冲激响应

Lathi 3.7-4 | 原书第 321 页

Boulet 第3章：Impulse Response of a Difference LTI System， p.109；Characteristic Polynomials and Stability， p.112

展开所引用3.7-3的要求：(a)判断系统阶数；(b)确定特征模态；(c)求单位冲激响应 h[n] 的闭式表达式。

3.7-4 Repeat Prob. 3.7-3 for a system described by the difference equation

$$y[n+3]-\frac{3}{10}y[n+2]-\frac{1}{10}y[n+1]=2x[n+1]$$

演算区

由特征根图判断离散系统模态与稳定性

Lathi 3.9-4 | 原书第 327 页

Boulet 第3章：Characteristic Polynomials and Stability of Differential and Difference Systems, p.112

同页跨栏的(a)-(e)及A-J共十幅原图全部附上；各图按比例绘制，单位圆为参考。

3.9-4 Figure P3.9-4 locates the characteristic roots of ten causal, LTID systems, labeled A through J. Each system has only two roots and is described using operator notation as $Q(E)y[n] = P(E)x[n]$. All plots are drawn to scale, with the unit circle shown for reference. For each of the following parts, identify all the answers that are correct.

(a) Identify all systems that are unstable.

(b) Assuming all systems have $P(E) = E^2$, identify all systems that are real. Recall that a real system always generates a real-valued response to a real-valued input.

(c) Identify all systems that support oscillatory natural modes.

(d) Identify all systems that have at least one mode whose envelop decays at a rate of 2^{-n}.

(e) Identify all systems that have only one mode.

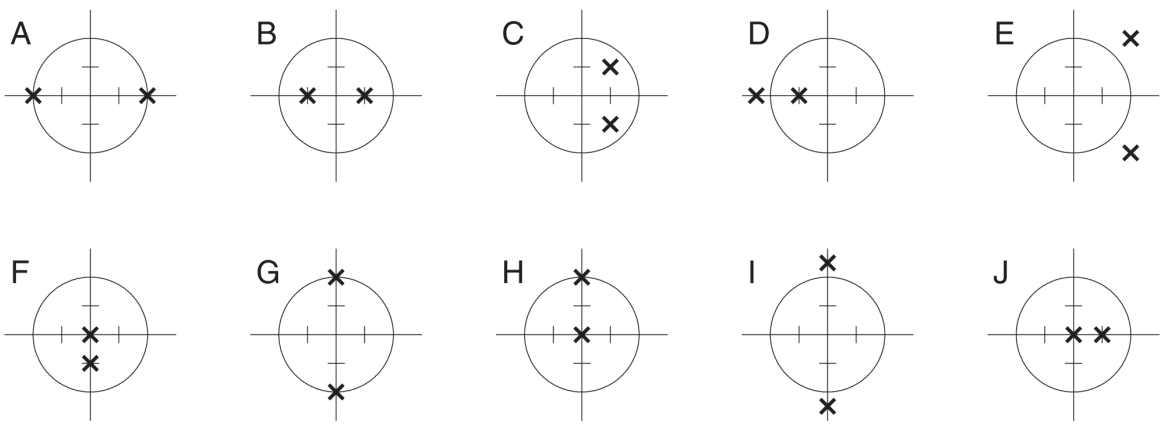


Figure P3.9-4

六种周期波形的三角傅里叶级数与幅相谱

Lathi 6.1-1 | 原书第 669、670 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal， p.134；Graph of the Fourier Series Coefficients: The Line Spectrum， p.137

所需原图 P6.1-1(a)至(f)六幅均已完整附上。本题要求紧凑三角式及其幅相谱。

6.1-1 For each of the periodic signals shown in Fig. P6.1-1, find the compact trigonometric Fourier series and sketch the amplitude and phase spectra. If either the sine or cosine terms are absent in the Fourier series, explain why.

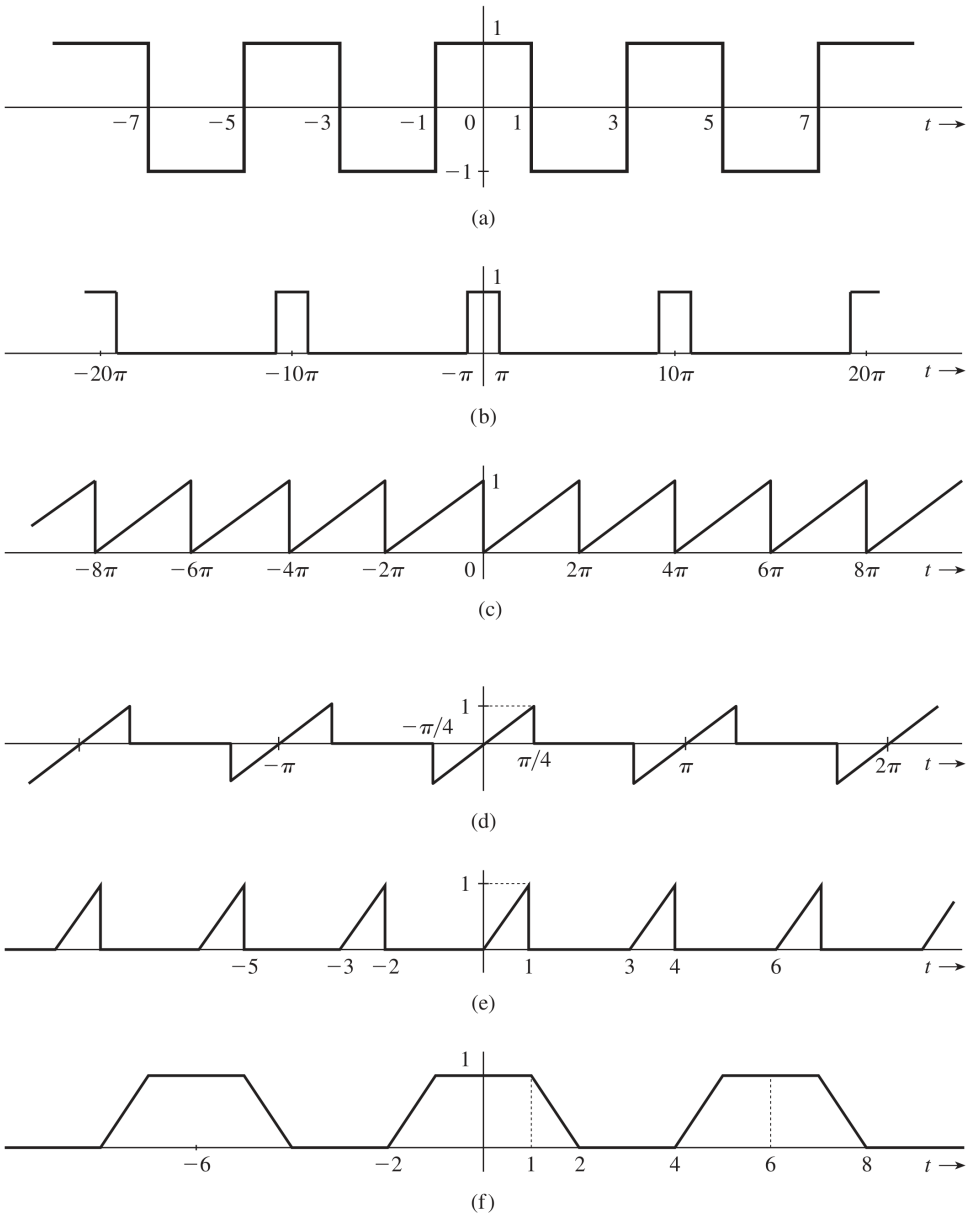


Figure P6.1-1

按基频、奇偶性与半波对称性构造周期延拓

Lathi 6.1-7 | 原书第 671、672 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials, p.132；Properties of Continuous-Time Fourier Series, p.139

原题跨栏、跨页，(a)至(f)及示意原图 P6.1-7均已附。展开提示中6.1-6的定义：半波对称是 x(t-T0/2)=-x(t)。只需画出满足条件的周期信号，不要求计算傅里叶系数。

6.1-7 Over a finite interval, a signal can be represented by more than one trigonometric (or exponential) Fourier series. For instance, if we wish to represent $x(t) = t$ over an interval $0 < t < 1$ by a Fourier series with fundamental frequency $\omega_0 = 2$, we can draw a pulse $x(t) = t$ over the interval $0 < t < 1$ and repeat the pulse every π seconds so that $T_0 = \pi$ and $\omega_0 = 2$ (Fig. P6.1-7a). If we want the fundamental frequency ω_0 to be 4, we repeat the pulse every $\pi/2$ seconds. If we want the series to contain only cosine terms with $\omega_0 = 2$, we construct a pulse $x(t) = |t|$ over $-1 < t < 1$, and repeat it every π seconds (Fig. P6.1-7b). The resulting signal is an even function with period π. Hence, its Fourier series will have only cosine terms with $\omega_0 = 2$. The resulting Fourier series represents $x(t) = t$ over $0 < t < 1$, as desired. We do not care what it represents outside this interval.

Sketch the periodic signal $x(t)$ such that $x(t) = t$ for $0 < t < 1$ and the Fourier series for $x(t)$ satisfies the following conditions.

(a) $\omega_0 = \pi/2$ and contains all harmonics, but cosine terms only

(b) $\omega_0 = 2$ and contains all harmonics, but sine terms only

(c) $\omega_0 = \pi/2$ and contains all harmonics, which are exclusively neither sine nor cosine

(d) $\omega_0 = 1$ and contains only odd harmonics and cosine terms

(e) $\omega_0 = \pi/2$ and contains only odd harmonics and sine terms

按基频、奇偶性与半波对称性构造周期延拓（续）

Lathi 6.1-7 | 原书第 671、672 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials, p.132；Properties of Continuous-Time Fourier Series, p.139

原题跨栏、跨页，(a)至(f)及示意原图 P6.1-7均已附。展开提示中6.1-6的定义：半波对称是 x(t-T0/2)=-x(t)。只需画出满足条件的周期信号，不要求计算傅里叶系数。

(f) $\omega_0 = 1$ and contains only odd harmonics, which are exclusively neither sine nor cosine.

[*Hint:* For parts (d), (e), and (f), you need to use half-wave symmetry discussed in Prob. 6.1-6. Cosine terms imply a possible dc component.] You are asked only to sketch the periodic signal $x(t)$ satisfying the given conditions. Do not find the values of the Fourier coefficients.

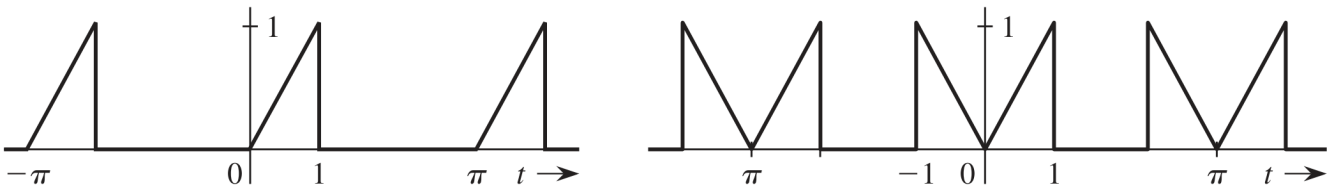


Figure P6.1-7

演算区

六种周期波形的复指数傅里叶级数与频谱

Lathi 6.3-1 | 原书第 672、670 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，p.134；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137

所需原图 P6.1-1(a)至(f)六幅均已完整附上；虽与6.1-1共用波形，本题要求复指数式及其频谱，是原书另一道独立习题。

6.3-1 For each of the periodic signals in Fig. P6.1-1, find exponential Fourier series and sketch the corresponding spectra.

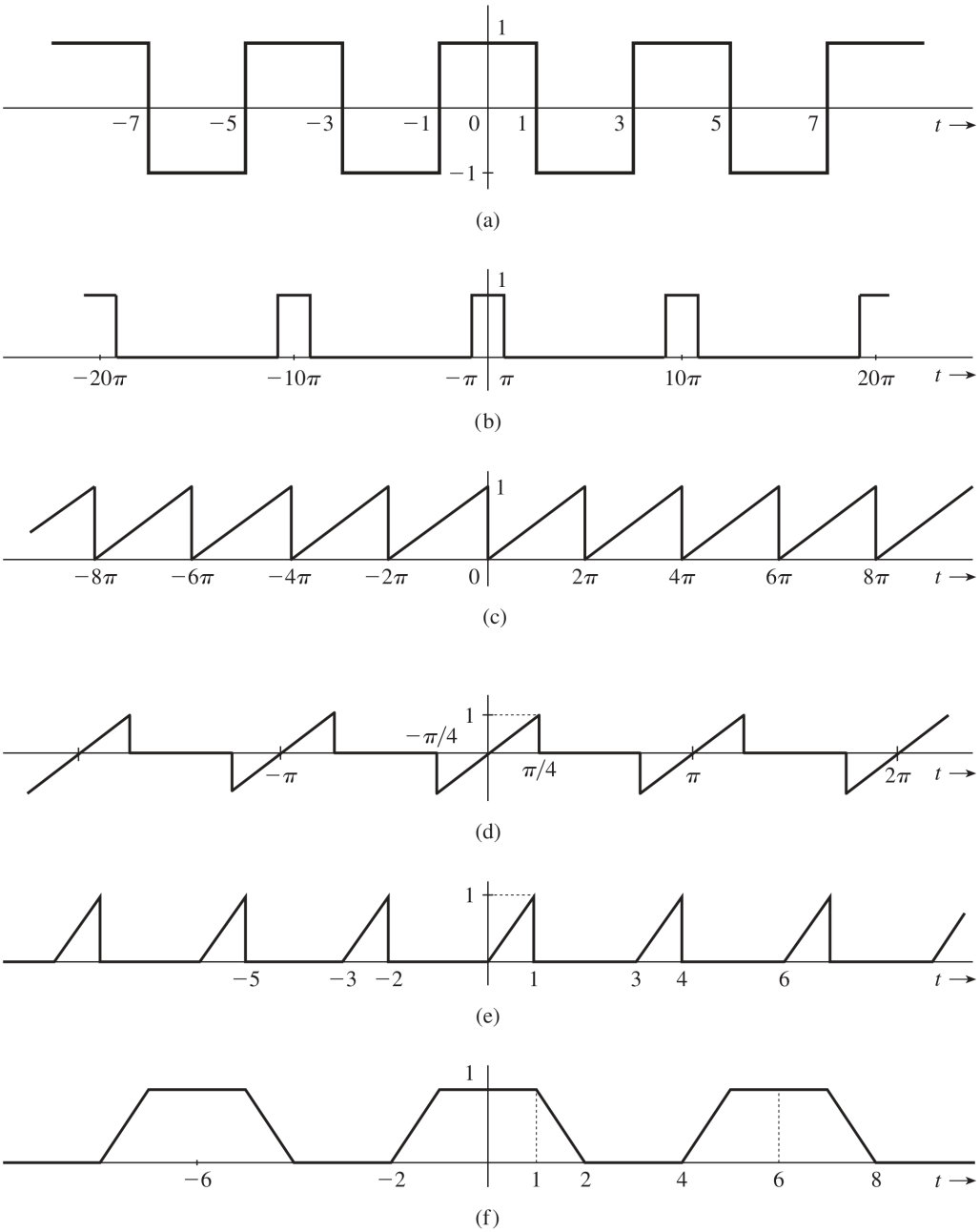


Figure P6.1-1

由误差平方推导向量最优投影系数

Lathi 6.5-1 | 原书第 676、643 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144

本题是傅里叶级数最优逼近的向量基础练习。原题所引用的式(6.32)已附，属于待推导命题，不是解答；其中x、y为实向量，y非零，c取使误差向量x-cy的长度平方最小的实数。

6.5-1 Derive Eq. (6.32) in an alternate way by observing that $\mathbf{e} = (\mathbf{x} - c\mathbf{y})$ and $|\mathbf{e}|^2 = (\mathbf{x} - c\mathbf{y}) \cdot (\mathbf{x} - c\mathbf{y}) = |\mathbf{x}|^2 + c^2|\mathbf{y}|^2 - 2c\mathbf{x} \cdot \mathbf{y}$.

$$c = \frac{\mathbf{x} \cdot \mathbf{y}}{\mathbf{y} \cdot \mathbf{y}} = \frac{1}{|\mathbf{y}|^2}\mathbf{x} \cdot \mathbf{y} \qquad (6.32)$$

演算区

最小误差逼近的正交性与方波正弦分量

Lathi 6.5-2 | 原书第 676、677、645 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144

原题跨页，(a)至(c)均已附。所需原图6.23已附，未附例题演算；(c)的比较区间为0<t<2π，图中方波在0<t<π为1、在π<t<2π为-1，比较信号为sin(t)。

6.5-2 A signal $x(t)$ is approximated in terms of a signal $y(t)$ over an interval (t_1, t_2):

$$x(t) \simeq cy(t) \qquad t_1 < t < t_2$$

where c is chosen to minimize the error energy.

(a) Show that $y(t)$ and the error $e(t) = x(t) - cy(t)$ are orthogonal over the interval (t_1, t_2).
(b) If possible, explain the result in terms of a signal-vector analogy.
(c) Verify this result for the square signal $x(t)$ in Fig. 6.23 and its approximation in terms of signal $\sin t$.

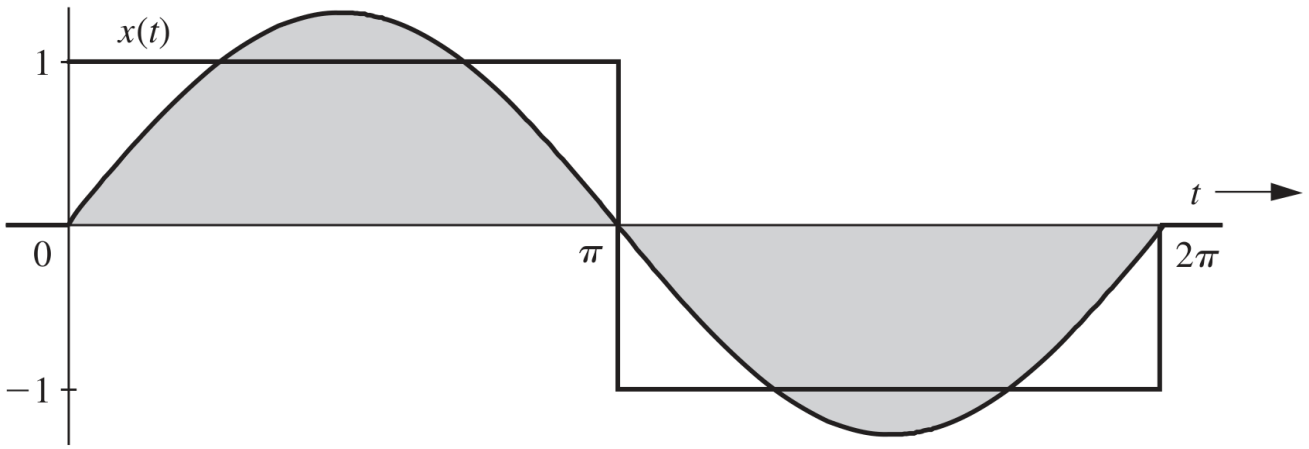


Figure 6.23 Approximation of a square wave in terms of a single sinusoid.